A THEORY OF THE LOW-FREQUENCY DIELECTRIC DISPERSION OF COLLOIDAL PARTICLES IN ELECTROLYTE SOLUTION[1,2]

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It is proposed that the low-frequency dielectric dispersion observed with colloidal particles suspended in electrolytes is due to the polarization of the counterion atmosphere around the particles by an external electric field. A simplified theory of this effect is developed for particles of spherical shape. The displacement of counterions in the double layer is shown to be equivalent to the existence of a surface capacitance displaying a diffusion-controlled relaxation. This effect can be expressed by an additional "apparent" dielectric constant of the suspended particles, exceeding their actual dielectric constant at low frequencies by many orders of magnitude. The theory is applied to the special case of particles suspended in a fairly well conducting electrolyte. Such colloidal systems display a particularly well pronounced low frequency dispersion effect. The results of previous experimental investigations with suspensions of uniformly sized polystyrene spheres are used in order to test the theory. Assuming reasonable values for the density and the mobility of the counterions in the double layer, the theory agrees very well with the static dielectric increment and the characteristic frequency of the dispersion as experimentally observed. A distribution of activation energies of the counterion motion on the particle surface is introduced to account for the relaxation spectrum of the dispersion curves.

Introduction

Suspensions of colloidal particles in aqueous electrolyte solutions generally display unusually high dielectric constants at low frequencies.[3] Thus, for sphere-like particles about 1 μ in diameter, the dielectric constant may rise well above 10^4 below 1 kc.[3,4] However, the experimental difficulties are considerable. Due to the appreciable electrolytic conductivity and the low frequencies used, it was possible only recently to achieve sufficient accuracy in the dielectric measurements and in the control of electrode polarization to take complete curves of the low frequency dielectric dispersion. For example, while investigating the dielectric behavior of biological material down to about 10 c.p.s., Schwan and co-workers found a well pronounced dispersion in the low frequency range.[5] This so-called α-dispersion perhaps is the same effect which occurs in the case of non-biological colloids. In fact, similar dispersion curves for fat particles and polystyrene spheres suspended in electrolyte solution have been measured by Schwan, *et al.*[3,5,6]

It has been suggested that displacements of electrical charges within the interface between the particle and the electrolyte may be responsible for the α-dispersion (*e.g.*, Murphy and Lowry[7] and Schwan[5]). Colloidal particles are electrically charged by fixed or adsorbed ions and are surrounded by counterions, forming an electric double layer or ionic atmosphere. A shift of ions within this layer by an external electrical field would result in a dielectric dispersion. Yet merely introducing an ohmic surface conductance around the particle into the Maxwell–Wagner theory of heterogeneous dielectrics,[8] as done by Miles and Robertson[9] and more recently by O'Konski,[10] cannot account for the experimental facts.[3] As has been demonstrated by Fricke and Curtis[4] and Schwan, *et al.*[3], only a frequency-dependent capacitive part of the surface admittance can explain the measured high dielectric constants. Unfortunately there has been no detailed justification of such a term so far.

Apparently it is possible to explain the effect only by a consideration of the microphysical structure of the double layer and the behavior of an individual charge carrier. Kirkwood and Shumaker[11] have considered this effect in the special case of the proton fluctuation around protein molecules by employing a statistical method. They demonstrated that the dielectric properties of protein solutions at radiofrequencies may be explained in this way just as well as by orientation of permanent dipoles, the latter being the commonly adopted interpretation.[12] In the general case of non-polar, spherical, and dielectrically-isotropic particles, an orientation mechanism is out of the question.[13] But a deformation of the ionic atmosphere due to an applied field can be anticipated and must be reflected in a corresponding dielectric relaxation. Whether this model explains the above-quoted dielectric phenomenon will be the subject of this paper.

Counterions on the surface of the highly-charged colloidal particle are strongly bound by electrostatic attraction. In order to escape from the surface into the free solution, they have to overcome a high potential barrier. Along the surface, however, they can be moved much more easily. Thus

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(2) This investigation was supported by National Institute of Health Grant H 1253 (C9).

(3) H. P. Schwan, G. Schwarz, J. Maczuk, and H. Pauly, *J. Phys. Chem.*, **66**, 2626 (1962).

(4) H. Fricke and H. J. Curtis, *ibid.*, **41**, 729 (1937). Further reference of older papers will be found here.

(5) H. P. Schwan, *Advan. Biol. Med. Phys.*, **5**, 147 (1957).

(6) H. P. Schwan and J. Maczuk, *Proc. Natl. Biophys. Conf., 1st, Columbus, Ohio*, 348 (1959).

(7) E. J. Murphy and H. H. Lowry, *J. Phys. Chem.*, **34**, 598 (1930).

(8) K. W. Wagner, *Arch. Elektrotech.*, **3**, 83 (1914).

(9) J. B. Miles, Jr., and H. P. Robertson, *Phys. Rev.*, **40**, 583 (1932).

(10) C. T. O'Konski, *J. Phys. Chem.*, **64**, 605 (1960).

(11) J. G. Kirkwood and J. B. Shumaker, *Proc. Natl. Acad. Sci. U. S.*, **38**, 855 (1952).

(12) J. L. Oncley, in Cohn and Edsall, "Proteins, Amino Acids and Peptides," Reinhold Publ. Corp., New York, N. Y., 1943.

(13) Moreover, an orientation mechanism would not account for the dielectric dispersion in question. The characteristic frequency of such an effect is inversely proportional to the third power of the diameter, according to Debye's well known equation of the relaxation time of oriented, spherical dipoles. This is in contrast to the experimental evidence discussed by Schwan, *et al.*[3] Furthermore, the quantitative value of such a characteristic frequency is found to be far below those of interest here.

they will be moved tangentially by an external field, polarizing the ion atmosphere and inducing an electric dipole moment of the particle. This concept has proved to be successful for rod-like particles in recent investigations on the orientation field effect of polyelectrolytes.[14-17] In place of a rigorous and difficult theory of the polarization of the counterion atmosphere, a fairly simplified model of the colloidal system is used in the calculations. Despite this rather crude model, the theoretical results yield a surprisingly good qualitative and quantitative representation of the experimental facts previously reported.[3]

Theory

Effect of an Electric Field on a Sphere with a Counterion Layer.—The colloidal particle will be introduced as a sphere of radius R and complex conductivity (in the presence of an alternating field of angular frequency ω)

$$K_i = \varkappa_i + i\omega\epsilon_r\epsilon_i \qquad (1)$$

suspended in an electrolyte solution of complex conductivity

$$K_a = \varkappa_a + i\omega\epsilon_r\epsilon_a \qquad (2)$$

where $\varkappa_i$, $\varkappa_a$ and ϵ_i, ϵ_a are the conductivities and the (relative) dielectric constants of the particle and the electrolyte, respectively. $\epsilon_r = 8.854 \times 10^{-12}$ f./m. is the absolute dielectric constant of free space. The sphere is electrically charged by fixed or adsorbed ions and is surrounded by counterions of electric charge e_0 and mechanical mobility u (*i.e.*, velocity per unit force), which can be moved only along the surface but not perpendicular to it. The electric double layer is assumed to be infinitely thin (in comparison with the radius of the particle) with a total net charge equal to zero. When the colloidal particle is not affected by external forces, the counterions are randomly distributed, so that the surface density σ (ions per unit area) is a constant σ_0. An applied external electric field E will result in a shift of the counterions along the surface, thus disturbing the random distribution. The counterion surface density may generally be denoted

$$\sigma = \sigma_0 + \bar{\sigma} \qquad (3)$$

where $\bar{\sigma}$ represents the counterion surface density in excess of the constant value σ_0.

It is suitable now to introduce spherical coordinates r and θ with $r = 0$ at the center of the sphere and $\theta = 0$ in the direction of E. If ψ_s is the electric potential on the surface of the sphere, the corresponding electric field

$$E_s = -\frac{1}{R} \times \frac{\partial\psi_s}{\partial\theta} \qquad (4)$$

causes a surface flux of counterions

$$j_E = -\frac{e_0\sigma\mu}{R} \times \frac{\partial\psi_s}{\partial\theta} \qquad (5)$$

In addition to the flux induced by the impressed electric field there is a counteracting diffusion flux which tends to re-establish the random distribution

$$j_D = -\frac{ukT}{R}\frac{\partial\sigma}{\partial\theta} \qquad (6)$$

Since the counterions can move only tangentially to the surface, any change of the counterion density must be due to those fluxes. Consequently the equation of continuity can be written as

$$\frac{\partial\sigma}{\partial t} - \text{div}\,(j_E + j_D) = 0 \qquad (7)$$

This is a partial differential equation for σ, namely

$$\frac{\partial\sigma}{\partial t} = \frac{ukT}{R^2} \times \frac{1}{\sin\theta} \times \frac{\partial}{\partial\theta}\left\{\sin\theta\,\frac{\partial\sigma}{\partial\theta} + \frac{e_0}{kT}\frac{\partial\psi_s}{\partial\theta}\,\sigma\sin\theta\right\} \qquad (8)$$

To compare the possible energy gain of a counterion responding to the external field with the energy due to thermal movement, the quantity

$$\gamma = \frac{e_0RE}{kT} \qquad (9)$$

may be introduced. For all cases discussed here

$$\gamma << 1 \qquad (10)$$

It stands to reason that the electric field along the surface is of the same order of magnitude as the external field or even smaller, therefore the factor of $\sigma \sin\theta$ in eq. 8 is also small compared with unity

$$\left|\frac{e_0}{kT} \times \frac{\partial\psi_s}{\partial\theta}\right| = \frac{e_0R}{kT} \cdot |E_s| \lesssim \gamma << 1 \qquad (11)$$

This indicates that the counterion density will deviate only little from the random distribution, that is

$$\sigma - \sigma_0 = \bar{\sigma} << \sigma_0 \qquad (12)$$

Because of relations 11 and 12, σ_0 can be substituted for σ in the second term on the right-hand side of eq. 8. Further simplification is possible if the external field is assumed to be alternating with the angular frequency ω, so that

$$E = E_0e^{i\omega t} \qquad (13)$$

Under these circumstances one may write

$$\frac{\partial\sigma}{\partial t} = \frac{\partial\bar{\sigma}}{\partial t} = i\omega\bar{\sigma} \qquad (14)$$

Using this result the equation for σ (8) takes the form

$$i\omega\bar{\sigma} = \frac{ukT}{R^2}\,\frac{1}{\sin\theta}\,\frac{d}{d\theta}\left\{\sin\theta\,\frac{d\bar{\sigma}}{d\theta} + \frac{e_0\sigma_0}{kT}\sin\theta\,\frac{d\psi_s}{d\theta}\right\} \qquad (15)$$

(14) G. Schwarz, *Z. Physik*, **145**, 563 (1956).
(15) M. Eigen and G. Schwarz, *J. Colloid Sci.*, **12**, 181 (1957).
(16) C. T. O'Konski and A. J. Haltner, *J. Am. Chem. Soc.*, **79**, 5634 (1957).
(17) G. Schwarz, *Z. physik. Chem.* (Frankfurt), **19**, 286 (1959).

The potential ψ_s must be derived from the over-all distribution of the electric potential. The potential in the electrolyte ψ_a as well as the potential inside the sphere ψ_i has to satisfy the Laplace equation

$$\Delta\psi_a = 0 \ (r > R) \tag{16}$$

$$\Delta\psi_i = 0 \ (r < R) \tag{17}$$

The boundary conditions are

1. $\lim_{r\to R} \psi_a = \lim_{r\to R} \psi_i = \psi_s$ (18)

2. $\psi_a \longrightarrow -Er\cos\theta$ for $r \longrightarrow \infty$ (19)

3. ψ_i must remain finite for $r \longrightarrow 0$ (20)

4. $K_a \frac{\partial\psi_a}{\partial r} - K_i \frac{\partial\psi_i}{\partial r} = -i\omega e_0\bar{\sigma}$ for $r \longrightarrow R$ (21)

The last condition results from the fact that the temporal change of the total surface density of charge depends completely upon the net influx of electrical charges perpendicular and parallel to the surface. The parallel flux is provided by the counterion motion, while the conductivities of the electrolyte and the particle are responsible for the flux in the perpendicular direction.

Now the three equations 15, 16, and 17 can be solved by standard methods. Using a representation of $\bar{\sigma}$ and ψ_s by means of Legendre polynomials $P_n(\cos\theta)$

$$\bar{\sigma} = \Sigma\, \alpha_n P_n(\cos\theta) \tag{22}$$

$$\psi_s = \Sigma\beta_n P_n(\cos\theta) \tag{23}$$

eq. 15 yields the following relation between the coefficients α_n and β_n of these series

$$\alpha_n = -\frac{1}{1 + i\omega\dfrac{R^2}{n(n+1)ukT}} \times \frac{e_0\sigma_0}{kT} \times \beta_n \tag{24}$$

As is well known the solutions of eq. 16 and 17 can be represented as

$$\psi_a = \Sigma\{A_{1n}r^n + B_{1n}r^{-n-1}\}P_n(\cos\theta) \tag{25}$$

and

$$\psi_i = \Sigma\{A_{2n}r^n + B_{2n}r^{-n-1}\}P_n(\cos\theta) \tag{26}$$

Taking into account the above-mentioned boundary conditions as well as eq. 22, 23, and 24 we find that all but the terms with $n = 1$ vanish. Under these circumstances eq. 24 is equivalent to

$$\bar{\sigma} = -\frac{1}{1 + i\omega\tau} \times \frac{e_0\sigma_0}{kT} \times \psi_s \tag{27}$$

where

$$\tau = \frac{R^2}{2ukT} \tag{28}$$

Thus we obtain a simple relaxation of the change in the counterion surface density. Obviously the relaxation time τ is determined by the rate of diffusion of the counterions along the particle. With respect to the actual electric potential ψ_s on the particle surface, the relaxation of the counterion layer is always diffusion-controlled. The same is not necessarily true with regard to the applied field, since there is in general a frequency-dependent phase shift between E_s and E.

According to eq. 5 and 6 the surface current density is

$$i_s = e_0(j_E + j_D) = -\frac{e_0^2 u\sigma}{R}\frac{\partial\psi_s}{\partial\theta} - \frac{e_0 ukT}{R} \times \frac{\partial\sigma}{\partial\theta} \tag{29}$$

In the first term σ_0 may be substituted for σ and in the second term $\bar{\sigma}$ for σ. Taking advantage of eq. 27

$$i_s = -\frac{i\omega\tau}{1 + i\omega\tau} \times \lambda_0 \frac{\partial\psi_s}{R\partial\theta} = \frac{i\omega\tau}{1 + i\omega\tau}\lambda_0 E_s \tag{30}$$

The quantity

$$\lambda_0 = e_0^2\, \sigma_0\, u \tag{31}$$

has been called surface conductivity by O'Konski,[10] who used the relation $i_s = \lambda_0 E_s$. As can be seen from eq. 30, this relation is indicated only at very very high frequencies or in cases where the diffusion can be neglected. In general the diffusion effect results in a phase shift between current and field on the surface. Thus λ_0 can be replaced by a complex surface conductivity

$$\lambda = \frac{i\omega\tau}{1 + i\omega\tau}\lambda_0 \tag{32}$$

It appears to be more suitable, however, to represent the current–field correlation by a surface dielectric constant ϵ_s^* instead of a surface conductivity, so that

$$\epsilon_s^* = \frac{1}{1 + i\omega\tau} \times \frac{\lambda_0\tau}{\epsilon_r} \tag{33}$$

The electric potentials ψ_a and ψ_i, as evaluated from the above-mentioned system of equations, can now be written

$$\psi_a = -Er\cos\theta + \frac{\bar{K}_i - K_a}{\bar{K}_i + 2K_a} R^3 E \frac{\cos\theta}{r^2} \ (r \geq R) \tag{34}$$

$$\psi_i = -\frac{3K_a}{2K_a + \bar{K}_i} Er\cos\theta \ (r \leq R) \tag{35}$$

where

$$\bar{K}_i = K_i + i\omega\epsilon_r \frac{2\epsilon_s^*}{R} \tag{36}$$

ψ_i becomes ψ_s by merely substituting R for r. Evidently the dielectric behavior of the sphere with the counterion layer is identical with that of a sphere without such a layer but an apparent dielectric constant

$$\bar{\epsilon}_i = \epsilon_i + \frac{2\epsilon_s^*}{R} \quad (37)$$

The additional dielectric constant

$$\Delta\epsilon_i = \frac{2\epsilon_s^*}{R} = \frac{1}{1 + i\omega\tau}\frac{e_0^2 R\sigma_0}{\epsilon_r kT} \quad (38)$$

may rise far above the actual dielectric constant ϵ_i.

The Dielectric Behavior of a Suspension of Spheres with a Counterion Layer.—If p is the volume concentration of spheres with a complex conductivity $\bar{K}_i$ suspended in material of complex conductivity K_a, the complex conductivity K of the system can be evaluated from the relation[3,18]

$$\frac{K - K_a}{K + 2K_a} = p\frac{\bar{K}_i - K_a}{\bar{K}_i + 2K_a} \quad (39)$$

This equation was essentially derived by Maxwell[19] for small values of p but has been found useful even for a fairly high volume concentration. The theoretical determination of K by means of eq. 39 can be eased if we assume a fairly well conducting electrolyte with

$$\varkappa_a >> \frac{\lambda_0}{R} \quad (40)$$

Since ϵ_i and $\varkappa_i$ are much smaller than ϵ_a and $\varkappa_a$, respectively, this relation is equivalent to

$$|\bar{K}_i| << |K_a| \quad (41)$$

Under these circumstances K follows from eq. 39 as

$$K = \frac{1 - p}{1 + \frac{p}{2}}K_a + \frac{9}{4}\frac{p}{\left(1 + \frac{p}{2}\right)^2}\bar{K}_i \quad (42)$$

This expression deviates from the complex conductivity K_∞ due to the simple Maxwell–Wagner effect only by the term $\bar{K}_i$ in place of K_i. Thus

$$K - K_\infty = \frac{9}{4}\frac{p}{\left(1 + \frac{p}{2}\right)^2}(\bar{K}_i - K_i) \quad (43)$$

is the additional complex conductivity resulting from the polarization of the counterion layer. It is suitable, however, to transform this relation into the corresponding equation for complex dielectric constants

$$\epsilon^* = \epsilon' - i\epsilon'' \quad (44)$$

where ϵ' is the real dielectric constant ϵ, and ϵ'' is the dielectric loss $\varkappa/\epsilon_r\omega$, κ being the ohmic conductivity. Taking into account eq. 36 and 38, we obtain for the complex dielectric increment

$$\epsilon^* - \epsilon_\infty^* = \Delta\epsilon^* = \frac{\Delta\epsilon_0}{1 + i\omega\tau} \quad (45)$$

with the static dielectric increment

$$\Delta\epsilon_0 = \frac{9}{4}\frac{p}{\left(1 + \frac{p}{2}\right)^2}\frac{e_0^2 R\sigma_0}{\epsilon_r kT} \quad (46)$$

$\Delta\epsilon^*$ displays a simple Debye-type dispersion and the characteristic frequency

$$f = \frac{1}{2\pi\tau} = \frac{ukT}{\pi R^2} \quad (47)$$

The counterion surface mobility u should be smaller than the mobility u_0 in free solution. In general the counterion motion along the surface is subject to an additional activation energy α, so that

$$u = u_0 e^{-\alpha/kT} \quad (48)$$

This activation energy α is of electrostatic origin. Since the fixed charges on the surface of the sphere are discretely distributed, the electric potential is changing periodically parallel to the surface. A counterion within the double layer must overcome potential barriers in order to follow an applied electric field. The height of the potential wall multiplied by the charge of a counterion yields the additional activation energy α. Generally the minimum distance δ between an individual counterion and its counter charge on the surface can be assumed to be at least several times smaller than the distance between the surface charges. In this case

$$\alpha \approx \frac{e_0^2}{\epsilon_a\delta} \quad (49)$$

For a δ of a few Angstrom units this equation yields an activation energy α of the order of magnitude of 1 kcal./mole.

Activation Energy Distribution and the Relaxation Spectrum.—On account of eq. 48, we can write the relaxation time τ (eq. 28 and 47) as

$$\tau = \tau_0 e^{\alpha/kT} \quad (50)$$

with

$$\tau_0 = \frac{R^2}{2u_0 kT} \quad (51)$$

If the suspended particles are not of uniform size, a relaxation spectrum according to the particle size distribution will occur. The resulting dispersion curves are flatter than those corresponding to a single relaxation time. However, as shown by experimental evidence, a relaxation spectrum may exist even in the case of particles of uniform size where all counterions are of the same kind. This discrepancy between our model theory and the experimental data simply may be due to the mathematical simplification of the problem. A more rigorous theoretical treatment of the diffusion-controlled relaxation of the counterion atmosphere possibly may result in a relaxation spectrum. Such an idea is suggested in view of the relaxation

(18) K. S. Cole, *J. Gen. Physiol.*, **12**, 29 (1928).

(19) J. C. Maxwell, "A Treatise on Electricity and Magnetism (1873)," Oxford Univ. Press, article 314.

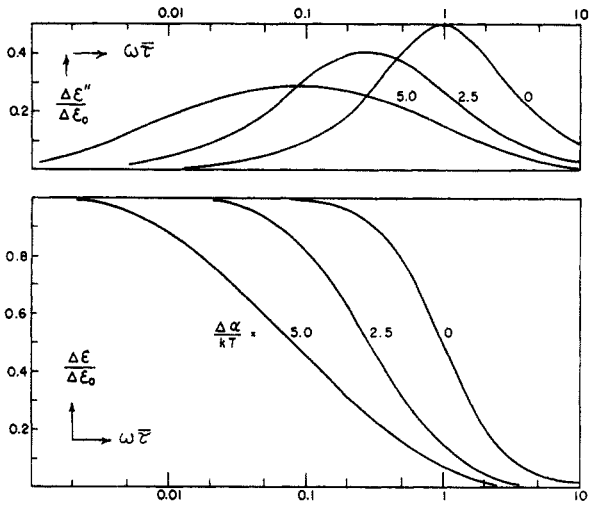


Fig. 1.—Effect of a statistical distribution of activation energies over a range $\Delta\alpha$ on the dispersion curves ($\bar{\tau}$ = relaxation time corresponding to the minimum activation energy $\bar{\alpha}$).

spectrum found for the ionic cloud of simple ions.[20,21] But the mechanism which is responsible for the relaxation spectrum will not be discussed here. Only a formal extension of the theory outlined above will be given in order to describe the experimental dispersion curves. This extension can only be done by a superposition of independent relaxation steps due to counterions with different activation energies, as is obvious from eq. 50 and 51. We define a distribution function $\rho(\alpha)$, so that there are in the mean

$$d\sigma_0 = \sigma_0\rho(\alpha)d\alpha \qquad (52)$$

counterions per unit area which are subject to an activation energy between α and $\alpha + d\alpha$. Then the relation

$$\int_0^\infty \rho(\alpha)d\alpha = 1 \qquad (53)$$

must hold. Now the total complex dielectric increment is derived simply from eq. 45 by integrating over the distribution of activation energies

$$\Delta\epsilon^* = \Delta\epsilon_0 \int_0^\infty \frac{\rho(\alpha)d\alpha}{1 + i\omega\tau_0 e^{\alpha/kT}} \qquad (54)$$

Since the static dielectric increment $\Delta\epsilon_0$ is not affected by the activation energy, it is unchanged (eq. 46). Naturally the function must be known in order to evaluate eq. 54 rigorously. However, the shape of the distribution curve is not very crucial with regard to the result of the integration. For example, it has been shown by Schwan[5] that several distribution functions which are symmetrical when plotted *vs.* ln τ (which is proportional to α in our case) yield almost identical dispersion curves. Hence for practical purposes we use a function $\rho(\alpha)$ which is a constant in an activation energy range from $\bar{\alpha}$ up to a maximum $\bar{\alpha} + \Delta\alpha$, but is zero otherwise. Because of eq. 53 the constant has to be $1/\Delta\alpha$. Then if

$$\xi = \epsilon^{-(\alpha-\bar{\alpha})/kT},\ \xi_0 = e^{-\Delta\alpha/kT} \qquad (55)$$

and

$$\bar{\tau} = \tau_0 e^{\bar{\alpha}/kT} \qquad (56)$$

eq. 54 gives

$$\Delta\epsilon^* = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \int_{\xi_0}^1 \frac{d\xi}{\xi + i\omega\bar{\tau}} = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \ln \frac{1 + i\omega\bar{\tau}}{\xi_0 + i\omega\bar{\tau}} \qquad (57)$$

Separating the real and the imaginary part of this expression, we obtain for the dielectric increment

$$\Delta\epsilon = \frac{\Delta\epsilon_0}{2 \ln \xi_0^{-1}} \ln \frac{1 + \omega^2\bar{\tau}^2}{\xi_0^2 + \omega^2\bar{\tau}^2} \qquad (58)$$

and for the increment of the dielectric loss

$$\Delta\epsilon'' = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \tan^{-1} \frac{(1 - \xi_0)\omega\bar{\tau}}{\xi_0 + \omega^2\bar{\tau}^2} \qquad (59)$$

Dispersion curves of this kind are shown in Fig. 1 for several values of $\Delta\alpha/kT$. With increasing width $\Delta\alpha$ of the activation energy range, they flatten out and differ more and more from the Debye-type curves ($\Delta\alpha = 0$). Simultaneously there is a decrease of the characteristic frequency f defined as the frequency for which $\Delta\epsilon = {}^1/_2\Delta\epsilon_0$. This behavior also is shown by the relation

$$f = \frac{1}{2\pi\bar{\tau}} \sqrt{\xi_0} = \frac{1}{2\pi\bar{\tau}} e^{-\Delta\alpha/2kT} \qquad (60)$$

as evaluated from eq. 58.

The maximum of the dielectric loss $\overline{\Delta\epsilon''}$ always is found at the characteristic frequency, but its absolute magnitude declines for increasing values of $\Delta\alpha$ while the shape of the curve becomes broader. The quantitative relation

$$\frac{2\overline{\Delta\epsilon''}}{\Delta\epsilon_0} = \frac{2 \tan^{-1} \frac{1 - \xi_0}{\sqrt{\xi_0}}}{\ln \xi_0^{-1}} \qquad (61)$$

as derived from eq. 58 and 59 is presented in Fig. 2 as a function of $\ln \xi_0^{-1} = \Delta\alpha/kT$. This curve can be used to determine $\Delta\alpha$ from measurements of $\Delta\epsilon$ and $\Delta\epsilon''$.

The quantity $\Delta\alpha$, which characterizes the width of the relaxation spectrum, can be determined even if only one of the dispersion curves (either that for $\Delta\epsilon$ or that for $\Delta\epsilon''$) has been measured. Since the degree of flattening depends on $\Delta\alpha$, one has to find out which value of ξ_0 in eq. 58 or 59, respectively, best fits the measured curve. For this purpose it is suitable to express $\bar{\tau}$, according to eq. 60, by a mean relaxation time

$$\tau_m = \frac{1}{2\pi f} = \frac{\bar{\tau}}{\sqrt{\xi_0}} \qquad (62)$$

because f and therefore τ_m can be obtained experimentally from either curve in a very simple way. Equations 58 and 59 then are written

(20) P. Debye and H. Falkenhagen, *Z. Elektrochem.*, **34**, 562 (1928).
(21) M. Eigen, *Discussions Faraday Soc.*, **24**, 24 (1957).

$$\Delta\epsilon = \frac{\Delta\epsilon_0}{2 \ln \xi_0^{-1}} \ln \frac{\xi_0^{-1} + \omega^2\tau_m^2}{\xi_0 + \omega^2\tau_m^2} \quad (63)$$

$$\Delta\epsilon'' = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \tan^{-1} \frac{1 - \xi_0}{\sqrt{\xi_0}} \frac{\omega\tau_m}{1 + \omega^2\tau_m^2} \quad (64)$$

Discussion

The theory presented in this paper has been developed on the basis of a model which is to a certain extent rather crude. Nevertheless the experimental data can be interpreted surprisingly well in the light of this theory. On the assumption of reasonable values for the basic parameters, *i.e.*, the counterion surface density σ_0 and the activation energy α, the theoretical results agree more than satisfactorily with the measurements.

Sufficient experimental data for a qualitative and quantitative test of the theoretical results can be found in the article of Schwan, *et al.*[3] The suspensions of polystyrene spheres used by these authors are most suitable for our purpose because the colloidal particles are non-polar spheres of extremely uniform and well known size. We shall discuss in particular the measurements carried out with polystyrene spheres in KCl solution. The diameters were either 1.17 or 0.557 μ. Experimental results for various concentrations of the polystyrene and the electrolyte, respectively, are shown in Fig. 8, 9, 10, and 11 of the mentioned article.

In order to apply the theory to a colloidal system, the electrolyte has to be fairly well conducting in accordance with relation 40. In fact, this inequality is generally satisfied in the case of the suspensions which are to be discussed. The quantity λ_0 can be determined from the conductivity of the system at higher frequencies (where $\lambda \rightarrow \lambda_0$ according to eq. 32). With regard to the previous investigation,[3] λ_0 has to be interpreted as the high frequency limit of the frequency-dependent part of the surface conductance (see footnote 23). It always has been found in the order of magnitude of 10^{-9} mho.[3,4] Moreover we can estimate λ_0 by means of eq. 31, using a result of Sieglaff and Mazur.[22] From electrophoretic measurements these authors determined the density of charge units on the surface of polystyrene spheres in a comparable system to be 1.9×10^{13} cm.$^{-2}$. Thus we may reasonably assume a σ_0 in the order of magnitude of 10^{13} cm.$^{-2}$. With potassium ions as the counterions, the calculation likewise yields a λ_0 around 10^{-9} mho. Hence for the particle sizes of interest, λ_0/R has to be expected to be in the order of magnitude of 10^{-5} mho/cm. This value is indeed well below the conductivity of the electrolyte solution in virtually all the suspensions in consideration.

Now we may compare the experimental results with the theoretical equations. Below about $p = 15\%$ the measured values of the static dielectric increment $\Delta\epsilon_0$ for the 1.17 μ sample, as shown in Fig. 10 of ref. 3, are in good agreement with eq. 46 if the counterion surface density is set as $\sigma_0 = 2.0 \times 10^{13}$ cm.$^{-2}$ for $\varkappa_a = 0.7$ mmho/cm. and $\sigma_0 = 1.5 \times 10^{13}$ cm.$^{-2}$ for $\varkappa_a$ 1.4 mmhos/cm. In view of the above remarks on λ_0 and σ_0, these

(22) C. L. Sieglaff and J. Mazur, *J. Colloid Sci.*, **15**, 437 (1960).

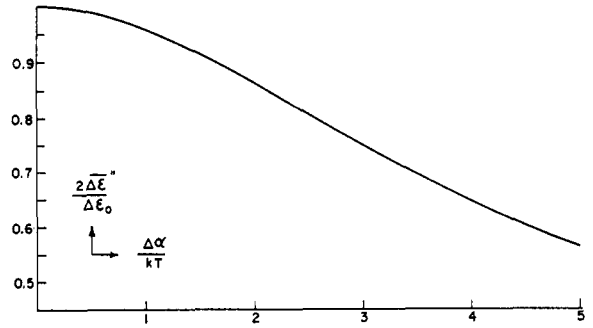


Fig. 2.—Effect of the activation energy distribution on the ratio of the maximum increment of the dielectric loss and the low-frequency limit of the dielectric increment.

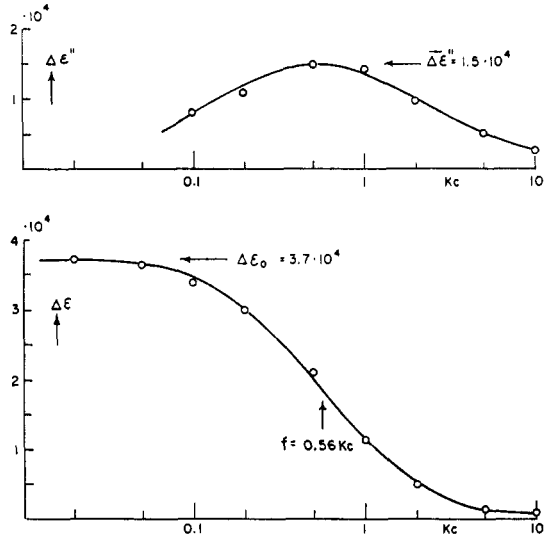


Fig. 3.—Low-frequency dielectric dispersion of $\Delta\epsilon$ and $\Delta\epsilon''$ as measured with a special sample: $2R = 1.17$ μ, $p = 22\%$, $K_a = 0.81$ mmho/cm. (KCl). The curves are computed theoretically with $\bar{\alpha} = 0$, $\Delta\alpha = 2.6$ kT.

σ_0 values obviously are very reasonable. At higher volume concentrations the measured dielectric increments exceed those computed from eq. 46 with the same σ_0 as before. However, since the basic eq. 39 has been proven only for small values of p, a disagreement between experiment and theory in this range of p is not surprising. Moreover, the quantity σ_0 may become a function of p here; thus the deviations between measurement and theory could be due also to an increase of σ_0 with increasing volume concentration. Such an interaction effect is not unlikely in view of the fact that other parameters of the suspension evidently affect the surface density of the counterions. This is shown with regard to the electrolyte concentration (conductivity) by the fact that it was necessary to use different values of σ_0 for different electrolyte conductivities. There are also indications of a slight dependence of σ_0 on the particle diameter, since according to Fig. 11 of ref. 3 the initial slope $\Delta\epsilon_0/p$ is not an exact linear function of R as it should be for a σ_0 independent of the diameter. Strictly speaking the counterion density σ_0 which appears in the theory represents only the ions bound to the particle surface, that is the "undissociated" counterions in contrast to the "dissociated" ones in free solution.[23] The degree

of "dissociation" and thereby σ_0 may well depend on the parameters κ_a, p, and R. Fortunately the previous assumption of a zero net charge of the double layer is not crucial here, so it may be dropped in compliance with this picture. An adequate extension of the basic theory will be discussed in a future paper.

An especially good agreement between our theory and the experiments is encountered in the case of the characteristic frequency f. Even the simple eq. 47 yields a proportionality to the inverse square of the radius of the particles, exactly as found experimentally (Fig. 9 of ref. 3). Furthermore it yields the quantitative results of the measurements by employing a counterion mobility $u = 1.35 \times 10^8$ c.g.s. units. Since the u_0-value for a potassium ion in water at 25° is 5×10^8 c.g.s. units, an activation energy $\alpha = 1.3\ kT = 0.8$ kcal./mole has to be assumed. This agrees very well with the estimation in the discussion of eq. 49. The successful interpretation of the characteristic frequency is not affected if an activation energy distribution is introduced in order to account for the shape of the dispersion curves. Then we have to use eq. 60 for the characteristic frequency f. The activation energy range $\Delta\alpha$ can be obtained from the measured ratio $2\overline{\Delta\epsilon}''/\Delta\epsilon_0 \approx 0.8$. According to Fig. 2 that results in $\Delta\alpha = 2.6\ kT$. With an $\bar{\alpha} = 0$, that is $\bar{\tau} = \tau_0$, eq. 60 results in the same quantitative relation as eq. 47. Dispersion curves computed on the assumption of such an activation energy distribution fit the experimental data very well as demonstrated by a set of measured curves in Fig. 3. According to eq. 49 the upper limit of the activation energy $\Delta\alpha = 2.6\ kT \approx 1.5$ kcal./mole corresponds to a minimum distance between opposite charges of $\delta = 3.0 \times 10^{-8}$ cm., obviously a very reasonable value.

In view of the very encouraging results obtained in comparing the experimental data with the theory, we may well conclude that the basic model of the colloidal particle used in this paper is essentially true. That means a diffusion-controlled polarization of a counterion layer around the colloidal particles is responsible for the low frequency dielectric dispersion under consideration. Particularly striking evidence for this conclusion is furnished by the following facts: (1) The characteristic frequency can be excellently expressed by the diffusion time of a counterion around the particle. (2) A very reasonable value of the counterion density on the surface of the particle results in a quantitative explanation of the increment of the dielectric dispersion.

(23) According to Cole,[24] Fricke and Curtis,[4] and Schwan, *et al.*,[3] the surface conductance of a colloidal particle is composed of two parts; one is frequency independent while the other one depends upon the frequency. The latter one apparently is due to the counterions which are "undissociated." The frequency independent part of the surface conductance causes virtually no dielectric dispersion in the cases of interest here[3] and is not discussed in this paper.

(24) K. S. Cole, *Cold Spring Harbor Symp. Quant. Biol.*, **1**, 1 (1933).

Outlook

The theory as presented in this paper has been applied only to a special type of colloidal system. However, there can be no doubt that the basic effect of the polarization of the counterion layer is of significance for all kinds of systems including those in the biological field, where ion layers on interfaces are involved. Obviously there are no essential difficulties in extending the theory to the more general case of suspensions of spherical particles with no restrictions concerning the electrical properties of the spheres and the surrounding medium. Furthermore, the theoretical considerations are suitable to be adapted to more complicated colloidal particles which may be shaped like ellipsoids or carrying a non-conductive shell. Particles of the latter kind are of special biological interest, since they represent the living cell with its cell wall. Investigations dealing with all these generalizations for the fundamental theory, as well as with the question of how the density σ_0 and the mobility u of the counterions in the double layer depend on the structure of the system, are in progress. They are to be discussed in future articles.

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